

The Plastic Detective

How AI Is Helping Protect Australia's Coastline

Original NAPLAN-style informational narrative | Year 6 Reading

Year 6 English

ACARA: AC9E6LY03

AC9E6LY05

AC9E6LY06

Name: _____

Date: _____

Score: ____ / 20

Learning Intention: I can analyse a text, infer meaning, evaluate evidence and justify my answers using details from the passage.

Reading Strategy

Read like a detective: notice clues, track evidence and think about the author's message.

Reading Passage

At 6:15 on a grey Tuesday morning, Dr Mira Patel stood on the boardwalk at Wattle Bay and watched a small drone rise above the water. The beach looked peaceful from the sand, but Mira knew the tide often carried hidden rubbish into the sheltered cove. Bottles, food wrappers and broken fishing line could collect between the rocks before anyone noticed.

Beside her, twelve-year-old Noah held a tablet computer and waited for the drone's first images to appear. Noah was visiting with his school science club, and he had expected a simple beach clean-up. Instead, he was watching a new system called ShoreScan, an artificial intelligence program designed to help protect Australia's coastline.

ShoreScan did not think like a person, but it could learn from patterns. Mira and her team had shown the program thousands of images of seaweed, shells, stones, birds, shadows and plastic. At first, it confused white shells with bottle caps and sea foam with floating bags. Each error helped the team improve it. Over time, ShoreScan became better at spotting shapes and colours linked to pollution.

When the drone flew over the eastern rocks, several orange dots appeared on Noah's screen. Each dot marked a place where the AI believed plastic waste might be trapped. Noah zoomed in and saw a clear drink bottle wedged in seaweed. Nearby, a thin loop of fishing line shimmered in the water.

'Why don't people just walk along the beach and pick it up?' Noah asked.

'They do,' Mira replied, 'and community clean-ups are still important. But people cannot check every rock pool, river mouth and stretch of coastline every day. AI helps us decide where to look first.'

The next part of the project used a floating robot no bigger than a laundry basket. It moved slowly through calm water, guided by maps created from the drone images. The robot collected small pieces of rubbish before they drifted into deeper water. It also recorded where each item was found, helping scientists understand how wind, rain and tides moved waste around the bay.

Mira explained that the goal was not to replace people. The robot could not persuade a family to use reusable containers, teach a shop to reduce packaging or convince a council to improve recycling bins. It was a tool, not a complete solution. The real solution required many choices: buying less single-use plastic, disposing of rubbish carefully, improving recycling systems and designing products that lasted longer.

By lunchtime, the team had collected two bottles, several wrappers, three pieces of fishing line and a handful of tiny plastic fragments. It was not a mountain of rubbish, but Mira looked pleased. A single loop of fishing line could trap a seabird's foot, and tiny fragments could be swallowed by fish.

Before the students left, Noah looked back at the drone resting in its case. He had arrived thinking technology would be the hero of the story. Now he understood something more interesting. Technology could be a powerful helper, but the coastline still depended on people noticing problems, changing habits and caring enough to act.

As the afternoon sun brightened the water, Noah wrote one sentence in his field notebook: 'The smartest tool is only useful when people choose to use it wisely.'

Comprehension Questions

Answer the questions after reading the passage. Use evidence where requested.

Section A

Multiple Choice

1. What was ShoreScan designed to do?

- ☐ A. Clean school classrooms
- ☐ B. Identify possible plastic waste along the coastline
- ☐ C. Replace all beach volunteers
- ☐ D. Teach students how to fly drones

2. Which statement best describes the main message of the passage?

- ☐ A. Drones are more important than people.
- ☐ B. Beaches no longer need clean-up teams.
- ☐ C. Technology can help, but people must also change their choices.
- ☐ D. Plastic pollution only affects large sea animals.

Section B

Find the Facts

3. Name two types of rubbish collected by the team at Wattle Bay.

4. What information did the floating robot record?

Section C

Vocabulary in Context

5. In the passage, what does durable mean?

- ☐ Easy to throw away
- ☐ Long-lasting
- ☐ Very colourful

Section D

Inference & Author's Purpose

6. Why did the author describe Noah changing his thinking?

Section E

Extended Response

7. Do you agree with Noah that technology is only useful when people use it wisely?

Use evidence from the passage to explain your answer in 4-6 sentences.

- ☐ Evidence used
- ☐ Key vocabulary
- ☐ Clear opinion
- ☐ Capital letters & full stops
- ☐ Answer checked

High Five! You're a Reading Comprehension Superstar!

☐ Vocabulary ☐ Inference ☐ Evidence Response - All done = Reader Award in Pet Town!

Teacher / Parent Feedback

Well done for...

Next step...

Star Rating: ☆☆☆☆☆

Teacher & Parent Guide

The Plastic Detective - Year 6 Reading Comprehension

Curriculum Alignment - ACARA v9.0

AC9E6LY03 - Analyse how text structures and language features work together to meet the purpose of a text.

AC9E6LY05 - Use comprehension strategies to interpret and analyse information and ideas.

AC9F6LY06 - Create written responses using evidence from texts.

Answer Key & Marking Guide

Question	Expected Response / Marking Guide	Marks
1	B - Identify possible plastic waste along the coastline.	2
2	C - Technology can help, but people must also change their choices.	2
3	Any two: bottles, wrappers, fishing line, tiny plastic fragments.	2
4	It recorded where each item was found.	2
5	Long-lasting.	2
6	To show that Noah learns technology is a helper, not the whole solution.	4
7	Strong responses include a clear opinion and evidence from the passage.	6

Differentiation Strategies

Support

Read passage aloud
Highlight key words
Answer Q1-Q4 first
Use sentence frames

At Level

Complete all questions
Use text evidence
Explain vocabulary
Self-check responses

Extension

Research a real clean-up tool
Write a balanced argument
Create an infographic
Compare another article

Teaching Tips & Home Activities

Teaching Tips: Preview vocabulary: durable, patterns, pollution, coastline, evidence. Discuss how the author uses Noah to show a change in understanding.

Home Activities: Do a family plastic audit, watch a short ocean clean-up video, debate whether technology or behaviour change matters more, or design a poster titled People + Technology = Cleaner Coasts.

Cross-Curriculum Links: Science - sustainability and ecosystems; Technologies - AI and data; HASS - community responsibility.